

In the Name of God, the Most Compassionate, the Most Merciful

**Faculty of Electrical and Computer Engineering
Isfahan University of Technology**

Reinforcement Learning - Second Semester 1401-1402

Assignment No. 2 - Due Saturday, 1401/3/6

Introduction to Multi-Step Methods and Combined Returns

1. Consider the Windy Gridworld introduced in Example 5-6 of the book. Solve this problem using the n-step Sarsa algorithm for $n = 1, 5, 10$.
2. Solve the above problem using the Sarsa(λ) algorithm for $\lambda = 0, 0.1, 0.5, 1$.
3. Compare and analyze the results obtained by these two methods. What is the effect of changing α on the two algorithms? Try, as far as possible, to illustrate the performance of these two algorithms with graphs; for example, the number of steps required to reach the goal across different trials, as well as any other graphs that you consider useful for a good analysis.

For the implementations, use Python. Do not display decimal values to more than three decimal places. (Always develop your programs incrementally, from small to large, and once you are confident that one part is correct, add another part to it.)

Submit the answers to the above questions and the results of your experiments in a report in PDF format, together with your programs.

Also submit a short video of your own execution of the program, together with an explanation of the experiments you performed.

Please pay attention to the following points:

- Assignments may be submitted up to two days after the stated deadline. A penalty of 10% of the grade will be deducted for each day of delay.
- Avoid colloquial writing and spelling errors, as they will result in a deduction of marks.
- Do not include photographs of handwritten work in your report; no credit will be given for them.
- Only assignments submitted through the course system will be graded. Under no circumstances should you email your assignment. Plan ahead so that your assignment is submitted through the system before the submission period ends.
- No credit will be given for assignments that are similar to others or have been downloaded.

Good luck!

Palhang